

**Problem Statement ID: SIH26033**

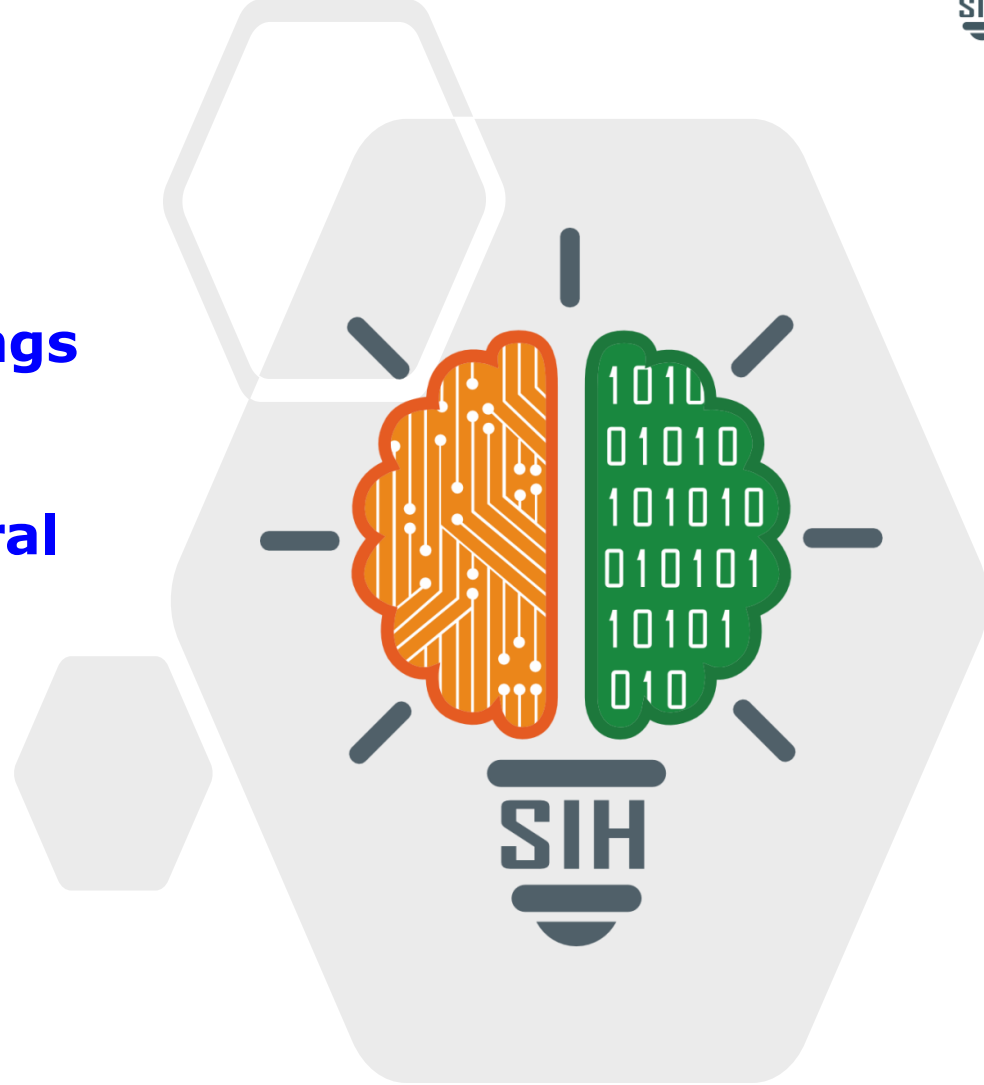
**Problem Statement Title: Multiple intermediaries reduce farmer earnings and increase consumer prices.**

**Theme: Agriculture, FoodTech & Rural Development**

**PS Category: Software**

**Team ID- 139407**

**Team Name: SINISTER SIX\***



## Understanding The Problem

- **Intermediate Fragmentation:** Produce goes through **4-6 layers of intermediaries** (aggregators, arhatiyas, and wholesalers), which drain out value without giving back any commensurate utility.
- **Low Farmer Realization:** Farmers realize **only 30-35%** of the final consumer price for perishables (proved by **studies by NABARD/ICAR**).
- **Price Inflation for Consumers:** Consumers have to **pay 2-3x the farm-gate price** due to the cumulative margins and inefficiencies in logistics systems.
- **Price Discovery Through Obscure Channels:** The perishable nature of produce creates vulnerabilities for farmers in **distress sales and market manipulation**.
- **Trust and Transport Bottleneck:** **Lack of reliable buyer channels** and transport logistics prevents **stranger-to-stranger** transactions.

## Our Solution

- **Direct Digital Marketplace:** **Links up farmers with buyers** without intermediaries through transparency in **e-commerce discovery**.
- **AI-powered Fair Pricing:** Intelligent **real-time market information system** avoids distressed selling and **ensures fair price**.
- **Verified Buyer Network:** Creates **trust and verification** channels for **safe transactions** of valuable, perishable goods.
- **Integrated Last-Mile Logistics:** Helps overcome major agri-tech challenges through **route optimization & order tracking**.
- **Modern Hybrid Mandi Coexistence:** Disintermediation of trade **without excluding traditional mandis** if so desired.
- **Vernacular & Rural-First Accessibility:** **Hindi + English support** and a **mobile-responsive UI** built for low-bandwidth rural networks

Live Prototype (60% Working)  
(KisanSetu-Agri-Marketplace)



### Key Innovations

**Role-Based Access**  
Separate Farmer & Buyer workspaces

**AI Fair Pricing**  
Obtain actual pricing range

**Route Optimization**  
TSP batching of farm pickups

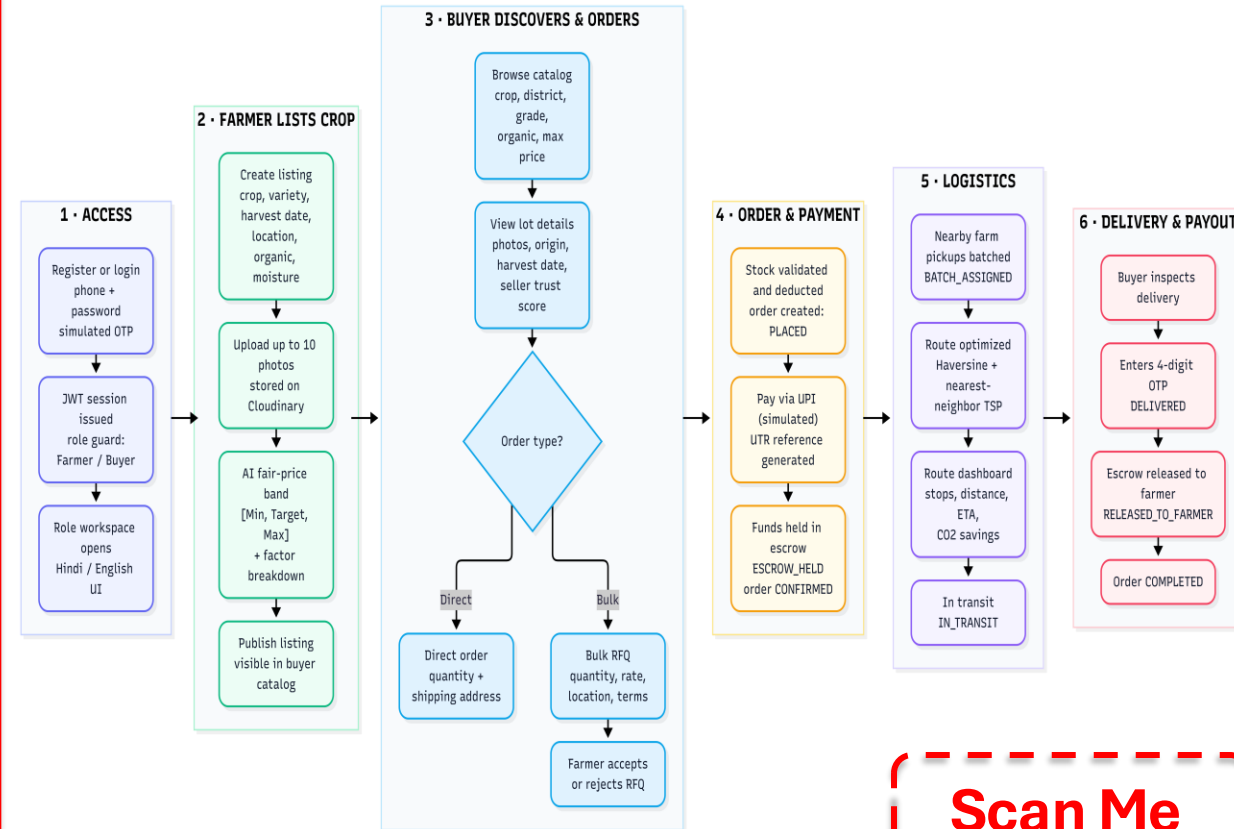
**Escrow & Dual-OTP**  
Money kept securely until OTP verification

**Trust Score**  
Build trust between farmers & buyers

## Technology Stack

- **Frontend** : **React 19, Vite 6, Tailwind CSS 4** mobile-first interface for Farmers and Buyers, in Hindi and English.
- **Backend**: **Python + FastAPI (Uvicorn, Pydantic v2) REST APIs** for auth, listings, photo uploads, pricing, orders, payments, RFQs, logistics and analytics.
- **Database**: **PostgreSQL / SQLAlchemy 2.0 + SQLite (default)** Store users, crops, orders.
- **AI / Machine Learning**: **Python + Scikit-learn ( NumPy, Pandas)**. Fair price band [Minimum, Target, Maximum] based on training on APMC mandi trading history.
- **Logistics**: **Haversine formula + TSP** Multi-farm batch pickup routing and an analysis.
- **Authentication & Security**: **JWT + bcrypt**, role-based access Phone/password login with simulated OTP
- **Deployment & Services**: **Vercel** (frontend + edge CDN) + **Render** (FastAPI backend) + **Neon** (PostgreSQL)
- **Supporting Services**: **Cloudinary** (image storage and CDN), **UptimeRobot** (keep-alive pings ), **GitHub Actions** (weekly encrypted database backups)

## Kisansetu-Workflow



Scan Me

Github Repo  
(Source Code)



## Feasibility of the Idea

- **Built on mature, widely used technologies** (React 19, FastAPI, PostgreSQL, scikit-learn)
- **Core modules implemented**: Farmer/Buyer workspaces (RBAC), AI fair-price engine, TSP route optimization, Hindi/English UI
- **Developed security framework**: JWT/bcrypt authentication, rate limiting, validation
- **Deployed on Vercel + Render + Neon** PostgreSQL, using Cloudinary for image storage
- **Mobile-first**, bilingual UI designed for low-bandwidth rural networks

## Practical value (platform analytics)

- Farmer price realization: **61.8%** vs. **32.4%** mandi baseline
- **19.5%** consumer savings, **4 of 5** intermediary layers bypassed
- Up to **33.5%** less transport distance through batched pickups

Calculations based on **(NABARD/ICAR Studies)**

## Challenge → Fix (Mitigation)

### 1. Digital Literacy → Simple, Mobile-First UX

- **Challenge (Farmer)**: Farmers with **limited digital literacy** may find a marketplace app hard to use.
- **Proposed fix**: Keep the **experience simple and mobile-first**, so a farmer can list produce and **see a fair price** without complex steps.

### 2. Buyer Trust → Verification + Ratings

- **Challenge (Buyer and Farmer)**: Buyers can't inspect produce or **judge seller reliability** remotely
- **Proposed fix**: **Verification badges** on farmer and buyer profiles, plus **ratings after completed orders**

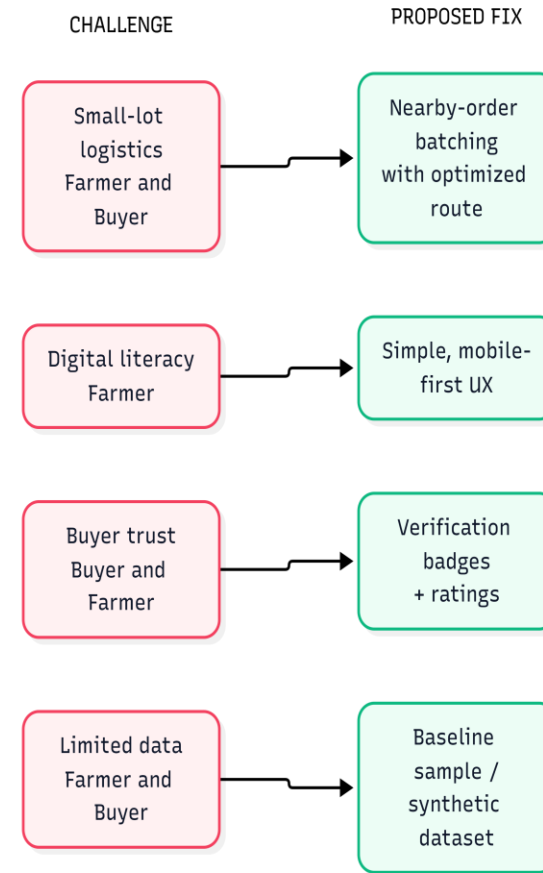
### 3. Small-Lot Logistics → Nearby-Order Batching

- **Challenge (Farmer and Buyer)**: **Small lots are costly to move** one by one, and **perishables lose value** in uncoordinated transit
- **Proposed fix**: **Group nearby orders** and farm pickups into one shared transit batch, with **stops sequenced by an optimized route**.

### 4. Limited Data → Baseline Sample / Synthetic Dataset

- **Challenge (Farmer and Buyer)**: A **new platform** has little transaction history, which can **weaken the fair-price band and benchmarks**.
- **Proposed fix**: Bootstrap the pricing engine with a **baseline sample/synthetic dataset**

## Challenges and Fixes



## Impacts on Framers

- **List Crops directly** with grade, harvest date, organic certification, packaging
- Get an AI fair price band **[Min, Target, Max] per quintal** while creating a listing
- See **mandi rate vs AI fair price vs retail price** side by side
- **Receive RFQs** and purchase requests **directly from wholesale buyers**

### Potential

- **Price realization of 61.8% vs 32.4%** in the traditional mandi baseline
- **Less distress selling**, through **AI fair pricing**
- Lower transit wastage through batched pickups (**~24.5% estimated spoilage reduction**)

## Impacts on Buyers (Consumers)

- Browse a **catalog filtered by crop, grade, district, organic certification and max price**
- View lot details: farm origin coordinates, **harvest date, seller trust score**, packaging
- Submit **bulk RFQs** and place farm-gate purchase orders
- Track orders through **PLACED → CONFIRMED → BATCH\_ASSIGNED → IN\_TRANSIT → DELIVERED → COMPLETED**

### Potential

- **19.5% consumer price savings**
- Payment released only after a **buyer-provided delivery OTP** on physical inspection (escrow is currently a simulation)

### Benefits

#### Social

- 4 of 5 layers cut\*
- Price transparency
- Hindi/English access
- Trust: score + OTP

#### Economic

- Farmer share 61.8% vs 32.4%
- *Consumer savings 19.5%*
- Quality-linked pricing
- Logistics km down 33.5%\*

### Impact

#### Farmers

##### Current

- Direct listing
- AI fair price
- Mandi vs AI vs retail view
- Direct buyer RFQs
- Hindi/English

##### Potential

- Price realization 61.8% vs 32.4%
- Less distress selling
- Spoilage down ~24.5%

#### Buyers

##### Current

- Filtered produce catalog
- Lot details + trust score
- Order status tracking

##### Potential

- Consumer savings 19.5%\*
- OTP-verified payment release (escrow simulated)

## Research

1. Examined **e-NAM for online agricultural trading**, price transparency, and connectivity between farmers and buyers.
2. Explored the model of **Farmer Producer Organisation (FPO)** for aggregation of small farmers and market accessibility.
3. Used AI agents for research (**Claude, Gemmni, Chatgpt**)
4. Examined **current agricultural marketplaces** in order to find out the **flaws in small lot aggregation, trust of buyers, logistics optimization, and price recommendation using AI.**
5. Reviewed government initiatives supporting **digital agriculture and direct market linkages.**

## References

1. e-NAM (National Agriculture Market), Government of India: <https://enam.gov.in>
2. Small Farmers' Agri-Business Consortium (SFAC), Ministry of Agriculture & Farmers' Welfare: <https://www.sfacindia.com>
3. Farmer Producer Organization(FPO) Scheme, SFAC: <https://www.sfacindia.com/FPOS.aspx>
4. NABARD, Farmer Producer Organizations & Agricultural Market Linkages:  
<https://www.nabard.org/contentsearch.aspx?AID=95&Key=ntro>
5. Ministry of Agriculture & Farmers' Welfare, Government of India